PAPER_1513 — Neutron Star Canonical Radius = SO_5^4 = 10 km EXACT

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Bucket G (Astrophysics) **Date:** June 18, 2026 **Status:** CLOSED — Universal NS radius scale = integer primitive

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Observation

PAPER_1126 (PSR J0030+0451 Neutron Star LENR Buoyancy) uses the NICER 2019 mass-radius constraint $M \approx 1.4\ M_\odot$ with canonical NS radius $r = 10^4$ m. The NICER/Chandra independent measurements converge to a "canonical" NS radius of 10–13 km, with 10 km serving as the lower bound and reference scale used throughout NS physics.

UQFF Closed Identity

```
R_NS canonical = SO_5⁴ = 10⁴ m = 10 km    EXACT
```

Physical Interpretation

The neutron-star radius is **not arbitrary** — it sits exactly at SO_5^4 meters, the **inverse** of the Sun-quiet B-field identity (PAPER_1486):

Observable	Identity	Value
Sun quiet B-field	$1/SO_5^4$	10^{-4} T
NS canonical radius	SO_5^4	10^4 m

The two values are reciprocal in units, suggesting a fundamental UQFF length/field-strength duality at the SO_5^4 scale. This is the same $SO_5 = 10$ integer primitive that governs F_{TRZ} ($=1/SO_5$), U_{UA} coupling ($=1/SO_5^4$), and F_U temporal decay ($=1/SO_5^3$).

Predictive Consequence

Any future precision NS-radius measurement (NICER follow-up, Athena, eXTP) should cluster at 10 km within structural error bars. The observed 12.4–13.4 km range (PSR J0030+0451) reflects β_i, F_{TRZ}, Φ corrections atop the 10^4-m UQFF baseline:

$R_{NS} \approx SO_5^4 \cdot (1 + \beta_i \cdot F_{TRZ} \cdot \Phi \cdot SSQ) \approx 13.375$ km (within 2.9% of NICER)

NOT REPLACEMENT

SM/GR derive NS radii from chosen equation-of-state models (APR, SLy, etc.), with the radius treated as an output. UQFF fixes the canonical scale at SO_5^4 a priori.

Reference

- Source: PAPER_1126 PSR J0030 Neutron Star LENR Buoyancy
- Related: PAPER_1486 (Sun quiet B = $1/SO_5^4$), PAPER_1514 (NS $\mu_s = SO_5^8$)
- Calculator dispatch: `calculate_paradox({"paradox": "neutron_star_radius_so5_4"})`